(e) Alcohols
Students should:
4.29C know that alcohols contain the functional group −OH
4.30C understand how to draw structural and displayed formulae for methanol, ethanol, propanol (*propan-1-ol only*) and butanol (*butan-1-ol only*), and name each compound ***the names propanol and butanol are acceptable***
4.31C know that ethanol can be oxidised by: • **burning in air or oxygen (complete combustion)** • **reaction with oxygen in the air to form ethanoic acid (microbial oxidation)** • **heating with potassium dichromate(VI) in dilute sulfuric acid to form ethanoic acid**
4.32C know that ethanol can be manufactured by: • **reacting ethene with steam in the presence of a phosphoric acid catalyst at a temperature of about 300 ºC and a pressure of about 60–70 atm** • **the fermentation of glucose, in the absence of air, at an optimum temperature of about 30 ºC and using the enzymes in yeast**
4.33C understand the reasons for fermentation, in the absence of air, and at an optimum temperature